

morphism between covering spaces of X is just a graph isomorphism that preserves the labeling and orientations of edges. Thus the covering spaces in (3) and (4) are isomorphic, but not by an isomorphism preserving basepoints, so the two subgroups of $\pi_1(X, x_0)$ corresponding to these covering spaces are distinct but conjugate. On the other hand, the two covering spaces in (5) and (6) are not isomorphic, though the graphs are homeomorphic, so the corresponding subgroups of $\pi_1(X, x_0)$ are isomorphic but not conjugate.

Some of the covering spaces (1)–(14) are more symmetric than others, where by a ‘symmetry’ we mean an automorphism of the graph preserving the labeling and orientations. The most symmetric covering spaces are those having symmetries taking any one vertex onto any other. The examples (1), (2), (5)–(8), and (11) are the ones with this property. We shall see that a covering space of X has maximal symmetry exactly when the corresponding subgroup of $\pi_1(X, x_0)$ is a normal subgroup, and in this case the symmetries form a group isomorphic to the quotient group of $\pi_1(X, x_0)$ by the normal subgroup. Since every group generated by two elements is a quotient group of $\mathbb{Z} * \mathbb{Z}$, this implies that every two-generator group is the symmetry group of some covering space of X .

Lifting Properties

Covering spaces are defined in fairly geometric terms, as maps $p: \tilde{X} \rightarrow X$ that are local homeomorphisms in a rather strong sense. But from the viewpoint of algebraic topology, the distinctive feature of covering spaces is their behavior with respect to lifting of maps. Recall the terminology from the proof of Theorem 1.7: A **lift** of a map $f: Y \rightarrow X$ is a map $\tilde{f}: Y \rightarrow \tilde{X}$ such that $p\tilde{f} = f$. We will describe three special lifting properties of covering spaces, and derive a few applications of these.

First we have the **homotopy lifting property**, or **covering homotopy property**, as it is sometimes called:

Proposition 1.30. *Given a covering space $p: \tilde{X} \rightarrow X$, a homotopy $f_t: Y \rightarrow X$, and a map $\tilde{f}_0: Y \rightarrow \tilde{X}$ lifting f_0 , then there exists a unique homotopy $\tilde{f}_t: Y \rightarrow \tilde{X}$ of $\tilde{f}_0$ that lifts f_t .*

Proof: For the covering space $p: \mathbb{R} \rightarrow S^1$ this is property (c) in the proof of Theorem 1.7, and the proof there applies to any covering space. $\square$

Taking Y to be a point gives the **path lifting property** for a covering space $p: \tilde{X} \rightarrow X$, which says that for each path $f: I \rightarrow X$ and each lift $\tilde{x}_0$ of the starting point $f(0) = x_0$ there is a unique path $\tilde{f}: I \rightarrow \tilde{X}$ lifting f starting at $\tilde{x}_0$. In particular, the uniqueness of lifts implies that every lift of a constant path is constant, but this could be deduced more simply from the fact that $p^{-1}(x_0)$ has the discrete topology, by the definition of a covering space.

Taking Y to be I , we see that every homotopy f_t of a path f_0 in X lifts to a homotopy $\tilde{f}_t$ of each lift $\tilde{f}_0$ of f_0 . The lifted homotopy $\tilde{f}_t$ is a homotopy of paths, fixing the endpoints, since as t varies each endpoint of $\tilde{f}_t$ traces out a path lifting a constant path, which must therefore be constant.

Here is a simple application:

Proposition 1.31. *The map $p_* : \pi_1(\tilde{X}, \tilde{x}_0) \rightarrow \pi_1(X, x_0)$ induced by a covering space $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ is injective. The image subgroup $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ in $\pi_1(X, x_0)$ consists of the homotopy classes of loops in X based at x_0 whose lifts to $\tilde{X}$ starting at $\tilde{x}_0$ are loops.*

Proof: An element of the kernel of p_* is represented by a loop $\tilde{f}_0 : I \rightarrow \tilde{X}$ with a homotopy $f_t : I \rightarrow X$ of $f_0 = p\tilde{f}_0$ to the trivial loop f_1 . By the remarks preceding the proposition, there is a lifted homotopy of loops $\tilde{f}_t$ starting with $\tilde{f}_0$ and ending with a constant loop. Hence $[\tilde{f}_0] = 0$ in $\pi_1(\tilde{X}, \tilde{x}_0)$ and p_* is injective.

For the second statement of the proposition, loops at x_0 lifting to loops at $\tilde{x}_0$ certainly represent elements of the image of $p_* : \pi_1(\tilde{X}, \tilde{x}_0) \rightarrow \pi_1(X, x_0)$. Conversely, a loop representing an element of the image of p_* is homotopic to a loop having such a lift, so by homotopy lifting, the loop itself must have such a lift. $\square$

If $p : \tilde{X} \rightarrow X$ is a covering space, then the cardinality of the set $p^{-1}(x)$ is locally constant over X . Hence if X is connected, this cardinality is constant as x ranges over all of X . It is called the number of **sheets** of the covering.

Proposition 1.32. *The number of sheets of a covering space $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ with X and $\tilde{X}$ path-connected equals the index of $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ in $\pi_1(X, x_0)$.*

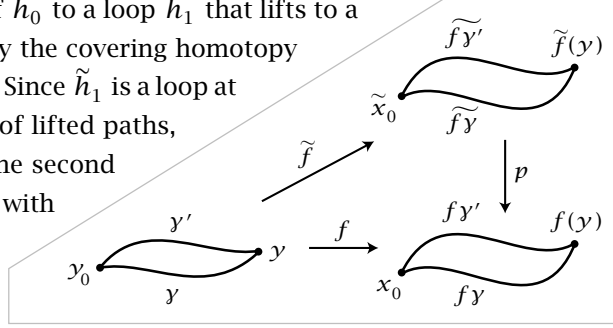
Proof: For a loop g in X based at x_0 , let $\tilde{g}$ be its lift to $\tilde{X}$ starting at $\tilde{x}_0$. A product $h \cdot g$ with $[h] \in H = p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ has the lift $\tilde{h} \cdot \tilde{g}$ ending at the same point as $\tilde{g}$ since $\tilde{h}$ is a loop. Thus we may define a function Φ from cosets $H[g]$ to $p^{-1}(x_0)$ by sending $H[g]$ to $\tilde{g}(1)$. The path-connectedness of $\tilde{X}$ implies that Φ is surjective since $\tilde{x}_0$ can be joined to any point in $p^{-1}(x_0)$ by a path $\tilde{g}$ projecting to a loop g at x_0 . To see that Φ is injective, observe that $\Phi(H[g_1]) = \Phi(H[g_2])$ implies that $g_1\bar{g}_2$ lifts to a loop in $\tilde{X}$ based at $\tilde{x}_0$, so $[g_1][g_2]^{-1} \in H$ and hence $H[g_1] = H[g_2]$. $\square$

It is important also to know about the existence and uniqueness of lifts of general maps, not just lifts of homotopies. For the existence question an answer is provided by the following **lifting criterion**:

Proposition 1.33. *Suppose given a covering space $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ and a map $f : (Y, y_0) \rightarrow (X, x_0)$ with Y path-connected and locally path-connected. Then a lift $\tilde{f} : (Y, y_0) \rightarrow (\tilde{X}, \tilde{x}_0)$ of f exists iff $f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(\tilde{X}, \tilde{x}_0))$.*

When we say a space has a certain property locally, such as being locally path-connected, we shall mean that each point has arbitrarily small open neighborhoods with this property. Thus for Y to be locally path-connected means that for each point $y \in Y$ and each neighborhood U of y there is an open neighborhood $V \subset U$ of y that is path-connected. Some authors weaken the requirement that V be path-connected to the condition that any two points in V be joinable by a path in U . This broader definition would work just as well for our purposes, necessitating only small adjustments in the proofs, but for simplicity we shall use the more restrictive definition.

Proof: The ‘only if’ statement is obvious since $f_* = p_* \tilde{f}_*$. For the converse, let $y \in Y$ and let γ be a path in Y from y_0 to y . The path $f\gamma$ in X starting at x_0 has a unique lift $\tilde{f}\gamma$ starting at $\tilde{x}_0$. Define $\tilde{f}(y) = \tilde{f}\gamma(1)$. To show this is well-defined, independent of the choice of γ , let γ' be another path from y_0 to y . Then $(f\gamma') \cdot (f\gamma)$ is a loop h_0 at x_0 with $[h_0] \in f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(\tilde{X}, \tilde{x}_0))$. This means there is a homotopy h_t of h_0 to a loop h_1 that lifts to a loop $\tilde{h}_1$ in $\tilde{X}$ based at $\tilde{x}_0$. Apply the covering homotopy property to h_t to get a lifting $\tilde{h}_t$. Since $\tilde{h}_1$ is a loop at $\tilde{x}_0$, so is $\tilde{h}_0$. By the uniqueness of lifted paths, the first half of $\tilde{h}_0$ is $\tilde{f}\gamma'$ and the second half is $\tilde{f}\gamma$ traversed backwards, with the common midpoint $\tilde{f}\gamma(1) = \tilde{f}\gamma'(1)$. This shows that $\tilde{f}$ is well-defined.



To see that $\tilde{f}$ is continuous, let $U \subset X$ be an open neighborhood of $f(y)$ having a lift $\tilde{U} \subset \tilde{X}$ containing $\tilde{f}(y)$ such that $p: \tilde{U} \rightarrow U$ is a homeomorphism. Choose a path-connected open neighborhood V of y with $f(V) \subset U$. For paths from y_0 to points $y' \in V$ we can take a fixed path γ from y_0 to y followed by paths η in V from y to the points y' . Then the paths $(f\gamma) \cdot (f\eta)$ in X have lifts $(\tilde{f}\gamma) \cdot (\tilde{f}\eta)$ where $\tilde{f}\eta = p^{-1}f\eta$ and $p^{-1}: U \rightarrow \tilde{U}$ is the inverse of $p: \tilde{U} \rightarrow U$. Thus $\tilde{f}(V) \subset \tilde{U}$ and $\tilde{f}|_V = p^{-1}f$, hence $\tilde{f}$ is continuous at y . $\square$

An example showing the necessity of the local path-connectedness assumption on Y is described in Exercise 7 at the end of this section.

Next we have the **unique lifting property**:

Proposition 1.34. *Given a covering space $p: \tilde{X} \rightarrow X$ and a map $f: Y \rightarrow X$ with two lifts $\tilde{f}_1, \tilde{f}_2: Y \rightarrow \tilde{X}$ that agree at one point of Y , then if Y is connected, these two lifts must agree on all of Y .*

Proof: For a point $y \in Y$, let U be an open neighborhood of $f(y)$ in X for which $p^{-1}(U)$ is a disjoint union of open sets $\tilde{U}_\alpha$ each mapped homeomorphically to U

by p , and let $\tilde{U}_1$ and $\tilde{U}_2$ be the $\tilde{U}_\alpha$'s containing $\tilde{f}_1(y)$ and $\tilde{f}_2(y)$, respectively. By continuity of $\tilde{f}_1$ and $\tilde{f}_2$ there is a neighborhood N of y mapped into $\tilde{U}_1$ by $\tilde{f}_1$ and into $\tilde{U}_2$ by $\tilde{f}_2$. If $\tilde{f}_1(y) \neq \tilde{f}_2(y)$ then $\tilde{U}_1 \neq \tilde{U}_2$, hence $\tilde{U}_1$ and $\tilde{U}_2$ are disjoint and $\tilde{f}_1 \neq \tilde{f}_2$ throughout the neighborhood N . On the other hand, if $\tilde{f}_1(y) = \tilde{f}_2(y)$ then $\tilde{U}_1 = \tilde{U}_2$ so $\tilde{f}_1 = \tilde{f}_2$ on N since $p\tilde{f}_1 = p\tilde{f}_2$ and p is injective on $\tilde{U}_1 = \tilde{U}_2$. Thus the set of points where $\tilde{f}_1$ and $\tilde{f}_2$ agree is both open and closed in Y . $\square$

The Classification of Covering Spaces

We consider next the problem of classifying all the different covering spaces of a fixed space X . Since the whole chapter is about paths, it should not be surprising that we will restrict attention to spaces X that are at least locally path-connected. Path-components of X are then the same as components, and for the purpose of classifying the covering spaces of X there is no loss in assuming that X is connected, or equivalently, path-connected. Local path-connectedness is inherited by covering spaces, so these too are connected iff they are path-connected. The main thrust of the classification will be the Galois correspondence between connected covering spaces of X and subgroups of $\pi_1(X)$, but when this is finished we will also describe a different method of classification that includes disconnected covering spaces as well.

The Galois correspondence arises from the function that assigns to each covering space $p: (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ the subgroup $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ of $\pi_1(X, x_0)$. First we consider whether this function is surjective. That is, we ask whether every subgroup of $\pi_1(X, x_0)$ is realized as $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ for some covering space $p: (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$. In particular we can ask whether the trivial subgroup is realized. Since p_* is always injective, this amounts to asking whether X has a simply-connected covering space. Answering this will take some work.

A necessary condition for X to have a simply-connected covering space is the following: Each point $x \in X$ has a neighborhood U such that the inclusion-induced map $\pi_1(U, x) \rightarrow \pi_1(X, x)$ is trivial; one says X is **semilocally simply-connected** if this holds. To see the necessity of this condition, suppose $p: \tilde{X} \rightarrow X$ is a covering space with $\tilde{X}$ simply-connected. Every point $x \in X$ has a neighborhood U having a lift $\tilde{U} \subset \tilde{X}$ projecting homeomorphically to U by p . Each loop in U lifts to a loop in $\tilde{U}$, and the lifted loop is nullhomotopic in $\tilde{X}$ since $\pi_1(\tilde{X}) = 0$. So, composing this nullhomotopy with p , the original loop in U is nullhomotopic in X .

A locally simply-connected space is certainly semilocally simply-connected. For example, CW complexes have the much stronger property of being locally contractible, as we show in the Appendix. An example of a space that is not semilocally simply-connected is the shrinking wedge of circles, the subspace $X \subset \mathbb{R}^2$ consisting of the circles of radius $1/n$ centered at the point $(1/n, 0)$ for $n = 1, 2, \dots$, introduced in Example 1.25. On the other hand, the cone $CX = (X \times I)/(X \times \{0\})$ is semilocally simply-connected since it is contractible, but it is not locally simply-connected.